

COM3021 – Data Science at Scale

Concept Drift Detection

DR YANDA MENG



Online learning

- There are three types of data shifts that will cause model decay
 - Covariate shifts
Changes in the input variables from the training data to the test data while the target variables are unchanged
 - Face recognition not being able to identify people wearing masks
 - Prior probability shifts
Changes in the target variables while input variables are unchanged
 - A model that could estimate the probability of someone having flu based on their symptoms during the COVID pandemic
 - Concept drifts
The relationship between input and output change over time in unforeseen ways
-

Concept drift

- Streaming data is dynamic and the underlying statistical characteristics can (and probably will) change over time
E.g. Consumer preferences and behaviours
 - **Gradual changes:** cultural or economic changes
 - **Abrupt changes:** Pandemic
 - **Recurring or cyclical changes:** seasons
 - Batch learner models will become outdated
-



Source: media.giphy.com

Drift detection

- There are many algorithms for drift detection
 - *ADaptive WINdowing method for concept drift detection (ADWIN)*
 - *Drift Detection Method (DDM)*
 - *Early Drift Detection Method (EDDM)*
-

ADWIN

- It maintains a window of variable size that adjusts itself in face of a drift
 - *It keeps the most recent data points in the stream and discards the older ones*
 - *It grows the window when no change is apparent*
 - *Shrinks it when data changes*

ADWIN: ADAPTIVE WINDOWING ALGORITHM

```
1  Initialize Window  $W$ 
2  for each  $t > 0$ 
3      do  $W \leftarrow W \cup \{x_t\}$  (i.e., add  $x_t$  to the head of  $W$ )
4      repeat Drop elements from the tail of  $W$ 
5          until  $|\hat{\mu}_{W_0} - \hat{\mu}_{W_1}| \geq \epsilon_{cut}$  holds
6          for every split of  $W$  into  $W = W_0 \cdot W_1$ 
7      output  $\hat{\mu}_W$ 
```

Figure 1: Algorithm ADWIN.

DDM

- DDM is used with other learning algorithms.
- For a given binary variable (True,False) representing an error (incorrect,correct). prediction, the instant error rate p_i (e.g., rate of incorrect predictions True) and standard deviation s_i at instant i are measured

p_{min} : The minimum recorded error rate

s_{min} : The minimum recorded standard deviation.

- if $p_i + s_i \geq p_{min} + 2 * s_{min}$ -> Warning zone
 - if $p_i + s_i \geq p_{min} + 3 * s_{min}$ -> Change detected
-